

# Syllabus

|            |                                         |
|------------|-----------------------------------------|
| Course     | CEVE 421 / 521: Climate Risk Management |
| Semester   | Spring 2026                             |
| Instructor | James Doss-Gollin                       |
| Email      | jdossgollin@rice.edu                    |
| Office     | Ryon 215                                |
| Class Days | MWF                                     |
| Class Time | 1-1:50 pm                               |

## 1 Overview

How should cities invest in flood protection when historical data no longer predicts the future? How do we weigh certain, immediate costs against uncertain future benefits or balance efficiency against equity?

These questions sit at the intersection of climate science, engineering, and decision-making under uncertainty. In practice, such decisions are often guided by intuition, politics, or precedent. This course focuses on the cases where rigorous quantitative analysis can do better.

The course follows a three-part arc:

1. **Risk Analysis:** Characterize climate hazards and quantify uncertainty using probabilistic methods including Monte Carlo simulation and extreme value theory.
2. **Decision Analysis:** Evaluate options under uncertainty using cost-benefit analysis, sensitivity analysis, and value of information.
3. **Optimization Under Uncertainty:** Find robust strategies using exploratory modeling, sequential decision-making, and multi-objective optimization.

Readings cover both foundational methods and real-world applications; programming labs let you implement and extend key concepts.

### 1.1 Prerequisites & Preparation

This is a 400-level CEVE course. In the past, many students from disciplines outside CEVE, including architecture, economics, and environmental science, have taken this class and done well. The following prerequisites are **required**:

- An introductory course in probability and statistics.
- Some exposure to Python, Julia, Matlab, R, or another programming language
- Standard undergraduate mathematics, including linear algebra and calculus (it's OK if these skills are a little bit rusty)

In addition, the following prerequisites are **encouraged**.

- Some exposure to Earth or environmental sciences, particularly regarding flooding
- Additional exposure (in a course or other experience) to probability theory, data analysis, machine learning, or statistics

If you are unsure whether your background gives you an adequate preparation for this course, **please contact the instructor!**

### What If My Skills Are Rusty?

If your programming, mathematics, or statistics skills are a little rusty, don't worry! We will review concepts and build skills over the course of the semester.

## 2 Course Objectives

By the end of this course, you will be able to:

1. **Risk Analysis:** Characterize climate hazards using probabilistic methods (Monte Carlo, Extreme Value Theory).
2. **Decision Analysis:** Critique and apply Cost-Benefit Analysis and Valuation frameworks.
3. **Optimization:** Apply robustness frameworks (XLRM) to find strategies that perform well across deep uncertainty.
4. **Integration:** Critique real-world climate plans by identifying hidden assumptions and ethical trade-offs.

## 3 A Community of Learning

### 3.1 Core Expectations

Course success involves a shared responsibility on the part of the instructor and the student.

As the instructor, **my responsibility** is to provide you with a structure and opportunity to learn. To this end, I commit to:

- provide organized and focused lectures, in-class activities, and assignments;
- encourage students to regularly evaluate and provide feedback on the course;
- manage the classroom atmosphere to promote learning;
- schedule sufficient out-of-class contact opportunities, such as office hours;
- allow adequate time for assignment completion;
- make lecture materials, class policies, activities, and assignments accessible to students.

Students are responsible for their own learning in the course. It is **your responsibility** to:

- attend all lectures;
- do all required preparatory work before class;
- actively participate in online and in-class discussions;
- begin assignments and other work early; and
- attend office hours as needed.

### 3.2 AI Policy

We approach AI tools (like ChatGPT, Claude, Copilot) as tools. There is no ban on AI in this course unless a specific assignment states otherwise. Instead, we operate on four principles:

1. **Shared Responsibility:** I commit to teaching you; you commit to learning. If you use AI to bypass thinking (e.g., generating code you don't understand), you cheat yourself of the skills you are spending your time (and money!) to learn.
2. **Assessment Design:** Most grading relies on in-class assessments (Exams & Quizzes) and critical critiques which are difficult to "game" with AI.
3. **Open Dialogue:** We will talk openly in class about how we are using these tools. We are learning together how to use them constructively. This requires being able to discuss our use of AI without worrying that we will be shamed for admitting that we use them. However, we commit to accepting constructive criticism of our (mis)use of AI tools

A separate document on AI is provided to help you use these technologies to deepen your understanding (e.g., as a tutor or debugger) rather than as a replacement for your own intellect.

### 3.3 Laptop Policy

To foster active discussion and deep cognitive presence, **laptops are not used during Monday Lectures and Wednesday Seminars**. Exceptions are made for tablet note-taking (flat on desk) or documented accommodations. Laptops are required for **Friday Lab Studios**.

### 3.4 Diversity and Inclusion

Our goal in this class is to foster an inclusive learning environment and make everyone feel comfortable in the classroom, regardless of social identity, background, and specific learning needs. As engineers, our work touches on many critical aspects of society, and questions of inclusion and social justice cannot be separated from considerations of systems analysis, objective selection, risk analysis, and trade-offs.

In all communications and interactions with each other, members of this class community (students and instructors) are expected to be respectful and inclusive. In this spirit, we ask all participants to:

- share their experiences, values, and beliefs;
- be open to and respectful of the views of others; and
- value each other's opinions and communicate in a respectful manner.

You are expected to be professional and courteous on all course interactions and platforms.

[Credit: Vivek Srikrishnan, Cornell]

**! Please, Be Excellent To Teach Other**

We all make mistakes in our communications with one another, both when speaking and listening. Be mindful of how spoken or written language might be misunderstood, and be aware that, for a variety of reasons, how others perceive your words and actions may not be exactly how you intended them. At the same time, it is also essential that we be respectful and interpret each other's comments and actions in good faith.

### 3.5 Accommodation for Students with Disabilities

If you have a documented disability or other condition that may affect academic performance you should: 1) make sure this documentation is on file with the Disability Resource Center (Allen Center, Room 111 / [adarice@rice.edu](mailto:adarice@rice.edu) / x5841) to determine the accommodations you need; and 2) talk with me to discuss your accommodation needs.

### 3.6 Accommodation for Scheduling Conflicts

If any of our class meetings conflict with your religious events, student athletics, or other non-negotiable scheduling conflict, please let me know ASAP so that we can make arrangements for you.

### 3.7 Getting Help

You can ask questions through Canvas Discussions and Office Hours.

1. **Do not** use email for questions about course content or labs, since other students may have related questions.
2. **Do** use email for questions about personal matters, such as scheduling conflicts, accommodations, etc.

### 3.8 Rice Honor Code

All students will be held to the standards of the Rice Honor Code, a code that you pledged to honor when you matriculated at this institution. If you are unfamiliar with the details of this code and how it is administered, you should consult the Honor System Handbook at [honor.rice.edu/honor-system-handbook/](http://honor.rice.edu/honor-system-handbook/). This handbook outlines the University's expectations for the integrity of your academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process.

## 4 Grading

Grades are weighted to reflect the different expectations for Undergraduate and Graduate students.

| Category      | CEVE 421 (Undergrad) | CEVE 521 (Grad) |
|---------------|----------------------|-----------------|
| Quizzes       | 20%                  | 15%             |
| Exams (3)     | 60% (20% each)       | 45% (15% each)  |
| Final Project | 20%                  | 20%             |

| Category           | CEVE 421 (Undergrad) | CEVE 521 (Grad) |
|--------------------|----------------------|-----------------|
| Seminar Leadership | N/A                  | 20%             |

## 4.1 Quizzes

Expect quizzes on Wednesdays. They serve three purposes:

1. **Theory Check:** Confirm you have engaged with the lectures and technical readings.
2. **Reading Check:** Confirm you have come to class ready to discuss the reading
3. **Lab Check:** Confirm you have completed the previous Friday's Lab.

These quizzes are not intended to be challenging if you have kept up with basic course expectations; they are primarily intended to make sure you are not falling behind. *The lowest two quizzes are dropped.*

## 4.2 Exams

There are three equally weighted exams. These are in-class exams designed to be completed in 50 minutes. Exams will contain a mix of multiple choice and open-ended questions.

## 4.3 Labs

Computational labs are for building your computational toolkit (Julia). Labs will be distributed through GitHub classroom. Solutions are posted immediately for self-correction.

Labs are not graded. However, Quizzes and Exams will test your ability to interpret and apply these computational methods. (A Wednesday quiz may contain questions about the previous Friday's lab).

### ! Hybrid

You may attend labs in-person or on Zoom. If there is an exam, lecture, or discussion on a Friday, you are expected to attend in-person.

## 4.4 Final Project

Working in small teams, you will act as expert consultants to audit a real-world climate planning document. Through two progressive Audit Memos, you'll analyze the plan's XLRM decision framework, evidence base, and robustness to uncertainty—culminating in an Executive Briefing and Written Report.

See Final Project for full details and timeline.

## 4.5 Graduate Student Requirements (CEVE 521)

In addition to the standard coursework, graduate students will **lead one Wednesday Paper Discussion session**. You will design an active learning experience—not simply present a paper.

See Graduate Seminar Requirements for full details.

## 5 Weekly Rhythm

| Day              | Activity   | Focus                               | Preparation                      |
|------------------|------------|-------------------------------------|----------------------------------|
| <b>Monday</b>    | Lecture    | Frameworks & Concepts               | See Preparation slide of lecture |
| <b>Wednesday</b> | Seminar    | <b>Quiz &amp; Active Discussion</b> | Complete Readings & Lab          |
| <b>Friday</b>    | Lab Studio | Coding & Debugging                  | Laptop required                  |

## 6 Preliminary Schedule

*Schedule is subject to change. Updated versions will be posted on the course website.*

| Week                                           | Dates            | Topic                                            | Lab / Exam                                                        | Project        |
|------------------------------------------------|------------------|--------------------------------------------------|-------------------------------------------------------------------|----------------|
| <b>Module 1 Risk Analysis</b>                  |                  |                                                  |                                                                   |                |
| <b>1</b>                                       | Jan 12, 14, 16   | Intro to Climate Risk Management                 | Lab 1: Julia Setup & Hello World                                  |                |
| <b>2</b>                                       | Jan 21, 23       | Hazard, Exposure, Vulnerability (XLRM)           | Lab 2: Hazard-Damage Convolution                                  |                |
| <b>3</b>                                       | Jan 26, 28, 30   | Climate Variability, Extremes, and Change        | Lab 3: Sea-Level Rise Scenarios and Uncertainty                   | Topic Proposal |
| <b>4</b>                                       | Feb 2, 4, 6      | Uncertainty, Monte Carlo & Risk Metrics          | Lab 4: Monte Carlo & Risk Metrics                                 |                |
| <b>Module 2 Decision Analysis</b>              |                  |                                                  |                                                                   |                |
| <b>5</b>                                       | Feb 9, 11        | Valuation & Benefit-Cost Analysis                |                                                                   |                |
| <b>6</b>                                       | Feb 16, 18, 20   | Optimal Static Decisions                         | Lab 5: Simulation-Optimization / <b>Exam 1</b> (Covers Weeks 1-4) |                |
| <b>7</b>                                       | Feb 23, 25, 27   | Sensitivity & Value of Information (Mon. Remote) | Lab 6: Value of Partial Information                               |                |
| <b>8</b>                                       | Mar 2, 4, 6      | Robustness & Deep Uncertainty                    | Lab 7: Robustness Metrics                                         |                |
| <b>9</b>                                       | Mar 9, 11, 13    | Trade-offs & Values (Mon. Workshop)              | <b>Exam 2</b> (Covers Weeks 5-8)                                  |                |
|                                                | March 16-20      | <i>Spring Break (No Class)</i>                   |                                                                   |                |
| <b>Module 3 Optimization Under Uncertainty</b> |                  |                                                  |                                                                   |                |
| <b>10</b>                                      | Mar 23, 25, 27   | Exploratory Modeling & Scenario Discovery        | Lab 8: Scenario Discovery                                         | Memo 1         |
| <b>11</b>                                      | Mar 30, Apr 1, 3 | Sequential Decision Making                       | Lab 9: Real Options & Dynamic Policy Search                       |                |
| <b>12</b>                                      | Apr 6, 8, 10     | Multi-Objective Optimization                     | Lab 10: Pareto Fronts                                             | Memo 2         |
| <b>13</b>                                      | Apr 13, 15, 17   | Case Studies                                     |                                                                   | Slides Due     |

| Week  | Dates          | Topic                       | Lab / Exam                                 | Project             |
|-------|----------------|-----------------------------|--------------------------------------------|---------------------|
| 14    | Apr 20, 22, 24 | Final Project Presentations |                                            |                     |
| Final | TBD            |                             | <b>Exam 3</b> ( <i>Covers Weeks 9-13</i> ) | Written Re-<br>port |

## Bibliography